Coordination compound

1. Which of the following statements are correctly defining salt?
(1) Product formed after reaction with Arrhenius acid & Arrhenius base.
(2) Compounds formed by the neutralization reaction of acid and base.
(3) Compounds which contains basic radical and acidic radical.
(4) Combination of cation and anion.
1) 1,2 and 3 only
2) 1,2 and 4 only
3) 1,2,3 and 4
4) 2,3 and 4 only

2. Which of the following is correct about simple salt?
1) Neutralization of single acids and single base.
2) Only are type of cation and only one type of anion.
3) NaCl is the example of simple salt.
4) All the above.

3. Assertion: Acidic salt is formed as a result of incomplete neutralization of acid.
Reason: Acidic salt must contain at least one H^+ ion.
1) Both Assertion and Reason are correct and the Reason is the correct explanation for Assertion.
2) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.
3) Assertion is true but reason is false.
4) Assertion is false but reason is true

4. Statement 1: Basic salt must contain at least one OH^- ion.
Statement 2: Basic slat is formed as a result of incomplete neutralization of base.
1) Statement-1 is true, Statement-2 is true and Statement-2 is correct explanation for Statement-I
2) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation for Statement-1
3) Statement-1 is true, Statement-2 is false
4) Statement-1 is false, Statement-2 is true

5. The part of radical which are derived from acid:
1) Cation
2) Anion
3) Acidic radical
4) (b) & (c) both

6. The part of complex salt which is present outside of coordination sphere is known as
1) Ionization sphere
2) Coordination sphere
3) Counter-ions which is ionizable in nature
4) Both a & c

7. Coordination sphere is nonionizable in nature because

1) In coordination sphere CMA and ligand form strong coordinate bond.
2) Coordination sphere is written in square brackets.
3) Coordination sphere is never dissociated into constituent's ions.
4) Coordination sphere is ionizable in nature

8. Central metal atom in coordination sphere has
1) Metal present inside coordination sphere in CMA.
2) CMA has vacant orbital which accept lone pair from ligand.
3) CMA has small size and high effective nuclear charge
4) All of the above

9. Mark the correct labelling of different terms used in coordination compounds:

$$[\underset{(iv)}{Co}(\underset{(i)}{NH_3})_{\underset{(iii)}{6}}]\underset{(ii)}{Cl_3}$$

1) (i) Central atom, (ii) Ionisation sphere, (iii) Coordination number, (iv) Ligands
2) (i) Ligands, (ii) Coordination number, (iii) Valency, (iv) Ionisation sphere
3) (i) Ionisation sphere, (ii) Ligands, (ii) Coordination number, (iv) Central atom
4) (i) Ligands, (ii) Ionisation sphere, (ii) Coordination number, (iv) Central atom

10. Which of the following is false for ligand?
1) It can be atom or molecule
2) Ligand can be cationic or anionic or neutral
3) Ligand act as a lewis base
4) Ligand cannot be cationic

11. Denticity is defined as:
1) Number of donor atoms present in a, ligand
2) Number of donor site in a ligand.
3) Number of co-ordinate bond formed by ligand
4) All of these

12. Which of the following is not example of monodentate ligands?
1) H_2O
2) Ethylene-diamine
3) Cl^-
4) CO

13. Which of the following is not example of bidentate ligands?
1) Diethylene triamine
2) Propylene diamine (pn)
3) Butylene diamine (bn)
4) Dimethyl glyoximato (DMG)

14. Which of the following statement is incorrect about DMG:-
1) It is tetradentate ligand
2) Chelating ligand
3) It is an anionic ligand
4) It is a symmetrical ligand

15. The donor atoms in EDTA are

1) Two N and Two O
2) Two N and four O
3) Four N and Two O
4) Three N and three O

16. The Oxidation state of Fe in brown ring complex $\left[Fe\left(H_2O\right)_5 NO\right]SO_4$ is

1) +1
2) +2
3) +3
4) +4

17. Consider:

Complex	Coordination no.
[A] $\left[CuCl_2\right]^-$	(i) 6
[B] $Ni\left(CO\right)_4$	(ii) 5
[C] $\left[PtCl_6\right]^{4-}$	(iii) 4
[D] $\left[CoCl_5\right]^-$	(iv) 2

Proper matching is:

1) A-i, B-ii, C-iii, D-iv
2) A-iii, B-iv, C-ii, D-iv
3) A-iv, B-iii, C-i, D-ii
4) A-i, B-iii, C-ii, D-iv

18. The oxidation number and coordination of chromium in the following complex is $\left[Cr\left(C_2O_4\right)_2\left(NH_3\right)_2\right]^{-1}$

1) $O.N. = +4,\ C.N. = 4$
2) $O.N. = +3,\ C.N. = 4$
3) $O.N. = -1,\ C.N. = 4$
4) $O.N. = +3,\ C.N. = 6$

19. The central metal atom or ion in a complex is identified by its coordination number, which is defined by

1) number of ligands around metal ion bonded by sigma bonds
2) number of ligands around metal ion bonded by pi-bonds
3) number of co-ordinate bonds metal atom/ion has accepted from ligands
4) the number of only anionic ligands bonded to the metal ion

20. The coordination number and oxidation number in complexes of $\left[Ag\left(NH_3\right)_4\right]\left[Ag\left(CN\right)_2\right]$ is

1) C.N. = 2, O.N. = +4 & C.N. = 4, O.N. = +4
2) C.N. = 2, O.N. = +1 & C.N. = 2, O.N. = +1
3) C.N. = 4, O.N. = +4 & C.N. = 4, O.N. = +1
4) C.N. = 2, O.N. = +4 & C.N. = 2, O.N. = +1

21. Trioxalatoaluminate (III) and tetrafluoro-borate (III) ions are:

1) $[AlC_2O_4]_3,[BF_4]^{3-}$
2) $[Al(C_2O_4)]^{3+},[BF_4]^{3+}$
3) $[Al(C_2O_4)]^{3-},[BF_4]^{-}$
4) $[Al(C_2O_4)]^{2-},[BF_4]^{2-}$

22. $K_3\left[Fe(CN)_6\right]$ is

(1) Potassium hexacynoferrous (III)
(2) Potassium hexacynoferrate (III)
(3) Potassium ferricyanide
(4) Hexaccynoferrate (III)

Correct answer is-

1) Only 1 and 2
2) Only 2 and 3
3) Only 1 and 3
4) Only 2 and 4

23. IUPAC name of $\left[Pt(NH_3)_3(Br)(NO_2)Cl\right]Cl$ is

1) Triamminechloridobromidonitroplatinum (IV) chloride
2) Triamminebromidonitrochloroplatinum (IV) chloride
3) Trimminebromidochloridonitroplatinum (IV) chloride
4) Triamminenitrochloridobromoplatinum (IV) chloride

24. The correct name of $\left[Pt(NH_3)_4Cl_2\right]\left[PtCl_4\right]$ is

1) Tetraammine dichloro platinum (IV) tetrachloroplatinate (II)
2) Dichloro tetra ammine platinium (IV) tetrachloro platinate (II)
3) Tetrachloro platinum (II) tetraammine platinate (IV)
4) Tetrachloro platinum (II) dichloro tetraamine platinate (IV)

25. The IUPAC name of $\left[Ni(NH_3)_4\right]\left[NiCl_4\right]$ is

1) Tetrachloronickel(II) tetraamminenickel(II)
2) Tetraamminenickel(II) tetrachloronickel(II)
3) Tetraamminenickel(II) tetrachloronickelate(II)
4) Tetraaminenickel(II) tetrachloridonickelate (IV)

26. In a complex, the correct statement is:

1) primary valency is ionizable
2) primary valency is non-ionisable
3) secondary valency is ionizable
4) all of these

27. The various hexacoordinate complexes' molecular formulas are listed below.

(1) $CrCl_3 \cdot 6NH_3$
(2) $CrCl_3 \cdot 5NH_3$
(3) $CrCl_3 \cdot 4NH_3$

If the number of NH3 ligands attached to central metal ion respectively are 6, 5 and 4, then the primary valencies in (1), (2) and (3) respectively are:

1) 3, 3, 3
2) 0, 1, 2
3) 3, 2, 1
4) 6, 5, 4

28. Which has the lowest molar electrical conductivity among the following?

1) $CoCl_3 \cdot 6NH_3$
2) $CoCl_3 \cdot 5NH_3$
3) $CoCl_3 \cdot 4NH_3$
4) $CoCl_3 \cdot 3NH_3$

29. When $AgNO_3$ is added to a solution of $Co(NH_3)_5Cl_3$, the precipitate of AgCl shows two ionizable chloride ions. This means –

1) Two chlorine atom satisfy primary valency and one chlorine atom satisfies primary valency as well as secondary valency.
2) One chlorine atom satisfies primary valency.
3) Two chlorine atoms satisfy secondary valency.
4) Three chlorine atoms satisfy secondary valency.

30. List the complex entities in the correct ascending order of electrical conductivity for aqueous solutions:

(I) $[Pt(NH_3)_6]Cl_4$ (II) $[Cr(NH_3)_6]Cl_3$ (III) $[Co(NH_3)_4Cl_2]Cl$

(IV) $K_2[PtCl_6]$

1) III < IV < II < I
2) IV < II < III < I
3) II < I < IV < III
4) I < II < IV < III

Answer Keys

1	3	**16**	1
2	4	**17**	3
3	2	**18**	4
4	1	**19**	3
5	4	**20**	2
6	4	**21**	3
7	1	**22**	2
8	4	**23**	3
9	4	**24**	1
10	4	**25**	3
11	4	**26**	1
12	2	**27**	1
13	1	**28**	4
14	1	**29**	1
15	2	**30**	1